CONSTRUCTIONS AND FINITE VERIFICATIONS FOR STRICT DISMANTLING LEVELS

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ABSTRACT. The recursive dismantling hierarchy measures how a graph can be reduced to one vertex by testing the graphs induced by successive neighbourhoods. This supplement supplies complete seed data and finite proofs for families of strict level k with $4k+4$ and $6k+6$ vertices. We relate constructions with two auxiliary vertices to the graph obtained by deleting edges between the exclusive parts of their profiles, and identify the one-edge completion that produces the eleven-vertex seed of the $3k+8$ family. Finally, we show that adding a layer shifts the reduced homology of every non-central neighbourhood by one degree. The accompanying calculations verify the finite hypotheses, clique counts and homological identities used in these arguments.

Repository: https://github.com/Geometric-and-Combinatorial-Structures/strictly-k-dismantlable-graphs
Release `v1.0.0`; file checksums in `SHA256SUMS.txt`.

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1. The constructions and their finite data

A graph is dismantled by deleting vertices until one vertex remains. In an ordinary dismantling, each deleted vertex is dominated by a neighbour. The recursive hierarchy permits a deletion at level k when the graph induced by its current neighbours is dismantlable at level $k-1$. A graph has *strict level* k if it admits a reduction at level k and none at level $k-1$.

The article *Linear-size graphs at every strict dismantling level* constructs graphs F_k and H_k of strict level k, for every $k \geq 1$, with

$$|V(F_k)| = 3k+8, \qquad |V(H_k)| = 4k+4.$$

The title, authors and current article reference are recorded in `ARTICLE.md`. This supplement develops the finite proof for H_k, gives a third family J_k of order $6k+6$, and explains the kernel completion and homological identities behind the layer construction. The two bounds from the article give the upper bound $\min\{4k+4, 3k+8\}$ on the least order at strict level k; they do not identify that least order in general.

Each family starts with a finite seed. A new layer contains a centre whose neighbourhood graph is the previous graph, together with auxiliary vertices that exclude first deletions at the lower level. To apply this construction, we need a reduction of the seed at its stated level, an obstruction to the preceding level, a reduction after deleting the first centre, and obstructions in the remaining

neighbourhoods. We supply the finite data for these tasks and prove why they suffice for every subsequent layer.

For example, H_1 has eight vertices. Its reduction at level one requires only two outer deletions, each justified by a single fold in a neighbourhood; the remaining graph is a cone. None of its original neighbourhoods is a cone, so no level-zero reduction can begin. Section 4 gives this example in full before treating the larger recursive certificates.

Organisation. Sections 2 and 3 fix the hierarchy and prove the general construction theorem. Section 4 supplies the $4k+4$ family, including its two prefixes; Section 5 checks a second application with five auxiliary types. Section 6 relates constructions with two auxiliary types to an edge-deleted kernel and identifies the completion producing F_1. Section 7 proves the degree shift in non-central neighbourhoods and the resulting homological criterion. Appendix A records the finite calculations and figures for the article's $3k+8$ family.

Conventions and notation. The symbols F_k and H_k retain their meanings in the article; J_k denotes the $6k+6$ family. Within a section, A denotes its seed, B its first extension and G_m the graph with m added layers. The seed vertex a_i has numerical label i in the data files. Homology is reduced homology over $\mathbb{F}_2$, with augmented chains in degree -1. The command following each finite calculation runs from the root of the accompanying repository and requires Python 3.11 or later.

2. The dismantling hierarchy

All graphs are finite, simple and undirected. For a graph H, write $V(H)$ and $E(H)$ for its vertex and edge sets and $H[S]$ for the subgraph induced by $S \subseteq V(H)$. In particular, $H - v = H[V(H) \setminus \{v\}]$. The *open and closed neighbourhood sets* of v are

$$N_H(v) = \{u \in V(H) : uv \in E(H)\}, \qquad N_H[v] = N_H(v) \cup \{v\}.$$

The *open neighbourhood graph* $H[N_H(v)]$ is the graph tested by the deletion rule.

A *cone* is a non-empty graph with a universal vertex, called an *apex*. The one-vertex graph K_1 is a cone, and the empty graph is not. A vertex u is *dominated* by $t \neq u$ if $N_H[u] \subseteq N_H[t]$. Deleting u under this condition is a *fold*, written $u \to t$. We can recognise a fold either from these closed neighbourhoods or by finding an apex in the open neighbourhood graph.

Lemma 2.1. *The graph $H[N_H(u)]$ is a cone if and only if u is dominated by another vertex.*

Proof. Suppose $N_H[u] \subseteq N_H[t]$ with $t \neq u$. Since u belongs to $N_H[t]$, the vertices u and t are adjacent. The same inclusion makes t adjacent to every vertex of $N_H(u) \setminus \{t\}$, so t is an apex of $H[N_H(u)]$. Conversely, if t is an apex of $H[N_H(u)]$, it is adjacent to u and to every other vertex of $N_H(u)$. Together with $t \in N_H[t]$, this gives $N_H[u] \subseteq N_H[t]$. For an isolated vertex the open neighbourhood graph is empty, and there is no vertex that dominates it. □

The cone test supplies the base of the recursive definition. At each higher level the same kind of test is applied to the graph on the neighbours, using the class from the preceding level.

Definition 2.2. *Let $\mathcal{D}_{-1}$ be the class of non-empty cones. For each integer $j \geq 0$, a non-empty graph H belongs to $\mathcal{D}_j$ if it can be reduced to K_1 by successive vertex deletions such that the open neighbourhood graph of each deleted vertex, in the graph remaining at that step, belongs to $\mathcal{D}_{j-1}$. A vertex whose open neighbourhood graph lies in $\mathcal{D}_{j-1}$ is j-dismantlable. A graph in $\mathcal{D}_j \setminus \mathcal{D}_{j-1}$ is strictly j-dismantlable. The empty graph belongs to none of these classes.*

This is the hierarchy of Fieux and Jouve [3], also used in [2]. By Lemma 2.1, the deletions allowed in $\mathcal{D}_0$ are folds, so $\mathcal{D}_0$ is the usual class of dismantlable graphs. At level one, the allowed deletions are the s-dismantling moves [1]. The graph K_1 belongs to every $\mathcal{D}_j$ because its reduction is empty. Membership is invariant under graph isomorphism. An isomorphism takes the apex of a cone to an apex and identifies the open neighbourhood graphs at each step of a reduction. Induction on j therefore preserves the admissibility of each deletion.

To compare two successive levels, we test the same deletion sequence at both levels.

Lemma 2.3. *The classes satisfy*

$$\mathcal{D}_{-1} \subseteq \mathcal{D}_0 \subseteq \mathcal{D}_1 \subseteq \mathcal{D}_2 \subseteq \cdots .$$

Proof. Start with a cone of apex z. Each other vertex can be folded onto z, which remains an apex as those vertices are deleted. This gives $\mathcal{D}_{-1} \subseteq \mathcal{D}_0$. Now suppose $\mathcal{D}_{j-1} \subseteq \mathcal{D}_j$ and $H \in \mathcal{D}_j$. Every neighbourhood graph tested in a level-j reduction of H belongs to $\mathcal{D}_{j-1}$ and hence to $\mathcal{D}_j$. The same sequence is therefore a level-$(j+1)$ reduction, completing the induction. □

For a graph with more than one vertex, a reduction must start with an admissible deletion. To use such a deletion to prove membership, we must also be able to reduce the graph it leaves.

Lemma 2.4. *Let* $|V(H)| > 1$ *and* $j \geq 0$.

(1) *If* $H[N_H(v)] \notin \mathcal{D}_{j-1}$ *for every* $v \in V(H)$, *then* $H \notin \mathcal{D}_j$.
(2) *If* $H[N_H(v)] \in \mathcal{D}_{j-1}$ *and* $H - v \in \mathcal{D}_j$ *for some* $v \in V(H)$, *then* $H \in \mathcal{D}_j$.

Proof. In the first case the neighbourhood test fails at every vertex, so no reduction can start. In the second, the test permits deleting v. The reduction of $H - v$ to K_1 at the same level then completes the sequence. □

For later use, abbreviate the neighbourhood graph by $L_H(v) = H[N_H(v)]$. For $|V(H)| > 1$ and $j \geq 0$, the first deletion and the remaining reduction give the equivalence

$$H \in \mathcal{D}_j \iff L_H(v) \in \mathcal{D}_{j-1} \text{ and } H - v \in \mathcal{D}_j \quad \text{for some } v \in V(H). \tag{1}$$

This equivalence will also justify the finite recursive certificates.

3. Adding layers and proving strictness

A seed at one strict level can produce examples at every higher level if each added layer provides an admissible first deletion and excludes all first deletions at the preceding level. We arrange for a centre to have the previous graph as its neighbourhood graph. The other vertices will be excluded by a non-zero alternating sum of clique counts. The same construction is used for all three families in this supplement.

3.1. **The layer construction.** Start with a non-empty graph A and fix an integer $d \geq 1$. Each layer has a centre and one auxiliary of each type in $\mathcal{T} = \{0, \ldots, d-1\}$. To specify the edges from the auxiliaries to the seed, choose a subset $P_j \subseteq V(A)$ for each $j \in \mathcal{T}$. We call P_j the *profile* of type j. Its vertices are the neighbours in A of every auxiliary of that type. To specify the edges between auxiliaries of distinct types, choose a graph R on $\mathcal{T}$.

Let G_m denote the graph obtained by adding m layers. Index the layers by $\mathcal{I}_m = \{0, \dots, m-1\}$ for $m \geq 1$, and put $\mathcal{I}_0 = \varnothing$. The vertices of G_m form the disjoint union

$$V(G_m) = V(A) \sqcup \{c_i : i \in \mathcal{I}_m\} \sqcup \{x_{j,i} : j \in \mathcal{T},\ i \in \mathcal{I}_m\}.$$

Here c_i is the centre of layer i, and $x_{j,i}$ is its auxiliary of type j. The following six rules specify every edge:

(1) The vertices of A induce the original graph A.
(2) The centres form a clique and are adjacent to every vertex of A.
(3) For each type j, its auxiliaries $x_{j,i}$, $i \in \mathcal{I}_m$, form a clique.
(4) For distinct types j, t, an auxiliary of type j is adjacent to an auxiliary of type t exactly when $jt \in E(R)$, regardless of their layers.
(5) The vertex $a \in V(A)$ is adjacent to $x_{j,i}$ exactly when $a \in P_j$.
(6) The centre c_i is adjacent to $x_{j,h}$ exactly when $i \neq h$.

The sixth rule is what makes a centre omit its own layer from its neighbourhood graph. For later use, write $B = G_1$, $p = c_0$ and $x_j = x_{j,0}$. Then $B[N_B(p)] = A$. The seed itself is $G_0 = A$. Adding a layer preserves all existing adjacencies. A vertex other than a centre is called *non-central*. Each added layer contributes $d + 1$ vertices, so

$$|V(G_m)| = |V(A)| + (d+1)m. \tag{2}$$

Lemma 3.1. *For $m \geq 1$ and $i \in \mathcal{I}_m$, the neighbourhood graph of c_i in G_m is isomorphic to G_{m-1}.*

Proof. The neighbours of c_i are the vertices of A and the vertices in layers $h \in \mathcal{I}_m$ with $h \neq i$:

$$N_{G_m}(c_i) = V(A) \sqcup \{c_h : h \neq i\} \sqcup \{x_{j,h} : j \in \mathcal{T},\ h \neq i\}.$$

Renumber the remaining layers by the increasing bijection $\rho_i : \mathcal{I}_m \setminus \{i\} \to \mathcal{I}_{m-1}$. Fix every vertex of A, send c_h to $c_{\rho_i(h)}$, and send $x_{j,h}$ to $x_{j,\rho_i(h)}$. The first five rules depend on the seed and on the roles or types of the vertices, which the map preserves. The sixth depends on whether the two layer indices agree. Since $h \neq t$ if and only if $\rho_i(h) \neq \rho_i(t)$, this rule is preserved as well. The map is therefore an isomorphism. For $m = 1$ it is the identity on A. □

To complete a reduction after deleting c_i, we also need to dismantle the graph that remains. We will fold each auxiliary outside layer i onto the auxiliary of the same type in layer i.

Lemma 3.2. *If $m \geq 2$, then $G_m - c_i \in \mathcal{D}_0$ for every $i \in \mathcal{I}_m$.*

Proof. Fix a type j. We will retain $x_{j,i}$ and fold every $x_{j,h}$ with $h \neq i$ onto it. The two auxiliaries have the same neighbours in A and among other types. Within their own type, their closed neighbourhoods agree because the copies form a clique. The only difference in their closed neighbourhoods in $G_m - c_i$ comes from the centres: $x_{j,i}$ is adjacent to all remaining centres, while $x_{j,h}$ is not adjacent to c_h. Hence

$$N_{G_m - c_i}[x_{j,h}] \subseteq N_{G_m - c_i}[x_{j,i}].$$

By Lemma 2.1, this inclusion permits the fold. We perform these folds for every type. The inclusions persist after earlier deletions, and all the targets are retained. The vertices left are those of A, the auxiliaries of layer i, and the $m - 1$ other centres. Each of these centres is adjacent to every other remaining vertex. Since $m \geq 2$, there is at least one such centre, so the remainder is a cone and folds to a single vertex. □

With only one layer, deleting p leaves no centre to serve as an apex. We will therefore put a separate hypothesis on $B - p$.

3.2. **An obstruction that changes sign.** To rule out a non-central vertex as a first deletion, we will check whether the clique complex of its neighbourhood graph has Euler characteristic different from one. We express this test through clique counts. A *clique* is a set of pairwise adjacent vertices. For a graph H, let $c_t(H)$ count its cliques of cardinality t, including the empty clique, so $c_0(H) = 1$. Set

$$C_H(z) = \sum_{t \geq 0} c_t(H) z^t, \qquad q(H) = C_H(-1).$$

The sum is finite, and $q(\varnothing) = 1$. In terms of the *clique complex* $\mathrm{Cl}(H)$, whose faces are the cliques of H,

$$q(H) = 1 - \chi(\mathrm{Cl}(H)) = -\widetilde{\chi}(\mathrm{Cl}(H)). \tag{3}$$

Here χ sums over the non-empty faces: a clique of cardinality $t \geq 1$ contributes $(-1)^{t-1}$. We use the reduced convention $\widetilde{\chi} = \chi - 1$, including $\widetilde{\chi}(\mathrm{Cl}(\varnothing)) = -1$. We will prove that q vanishes on every dismantling class, so a non-zero value in a neighbourhood graph rules out deleting its vertex. The vertex-deletion recurrence for the *clique polynomial* [4] lets us follow q through a reduction.

Lemma 3.3. *For every vertex v of a graph H,*

$$C_H(z) = C_{H-v}(z) + z C_{H[N_H(v)]}(z), \qquad q(H) = q(H - v) - q(H[N_H(v)]).$$

A cone has $q = 0$, and a fold preserves q.

Proof. Partition the cliques of H according to whether they contain v. Those that omit v are exactly the cliques of $H - v$. Removing v from each clique that contains it gives a clique of $H[N_H(v)]$, and adjoining v reverses the operation. Since adjoining v increases the cardinality by one, these cliques contribute the term $z C_{H[N_H(v)]}(z)$ in the polynomial identity. Evaluating at $z = -1$ gives the recurrence for q.

In a cone, pair each clique that omits the apex with the clique obtained by adjoining the apex. The paired contributions to q cancel. By Lemma 2.1, a vertex v deleted by a fold has a cone as its neighbourhood graph. The recurrence thus gives $q(H) = q(H - v)$ for that deletion. □

Taking coefficients in the polynomial recurrence gives

$$c_t(H) = c_t(H - v) + c_{t-1}(H[N_H(v)]) \qquad (t \geq 1). \tag{4}$$

The initial values are $c_0(H) = 1$ and $c_t(\varnothing) = 0$ for $t \geq 1$. Both graphs on the right have fewer vertices, so this determines every coefficient.

The cone calculation provides the base case for an induction on the dismantling level.

Lemma 3.4. *If $H \in \mathcal{D}_r$ for an integer $r \geq -1$, then $q(H) = 0$. Consequently, $q(H) \neq 0$ excludes H from every such class.*

Proof. The case $r = -1$ follows from Lemma 3.3, since $\mathcal{D}_{-1}$ consists of cones. For $r \geq 0$, suppose that q vanishes on $\mathcal{D}_{r-1}$ and take a reduction of H at level r. At each deletion, the neighbourhood graph in the current graph belongs to $\mathcal{D}_{r-1}$. Its value of q is zero, so the recurrence shows that the deletion preserves q. The reduction ends at K_1, with $q(K_1) = 1 - 1 = 0$. Hence $q(H) = 0$. □

A zero value of q leaves membership undecided. To exclude the non-central vertices we need non-zero values on their neighbourhood graphs. It suffices to compute q on the neighbourhood graphs of the seed vertices and the auxiliaries in the first extension B, because adding a layer changes the sign of the corresponding values. For $a \in V(A)$ and $j \in \mathcal{T}$, write

$$\eta_a = q(B[N_B(a)]), \qquad \theta_j = q(B[N_B(x_j)]).$$

Lemma 3.5. *For arbitrary seed, profiles and type graph, and every $m \geq 1$,*

$$q(G_m[N_{G_m}(a)]) = (-1)^{m-1}\eta_a \qquad (a \in V(A)), \tag{5}$$

$$q(G_m[N_{G_m}(x_{j,i})]) = (-1)^{m-1}\theta_j \qquad (j \in \mathcal{T},\ i \in \mathcal{I}_m). \tag{6}$$

Proof. We argue by induction on m. For $m = 1$, both assertions follow from the definitions of η_a and θ_j. Let $m \geq 2$ and assume the formulas hold at $m-1$. In each case H is the neighbourhood graph of a non-central vertex. We apply the recurrence inside H to a centre of G_m that lies in H. We choose that centre so that deleting it leaves a graph with $q = 0$. The remaining term in the recurrence is then the negative of q on the centre's neighbourhood graph in H.

First fix $a \in V(A)$ and put $H = G_m[N_{G_m}(a)]$. Its vertices are the neighbours of a in A, all the centres, and all copies of each type j with $a \in P_j$. We may choose any layer i, since H contains every centre. In $H - c_i$, fold $x_{j,h}$ onto $x_{j,i}$ for each of the types present and every $h \neq i$. The closed-neighbourhood inclusions from the proof of Lemma 3.2 still hold upon restriction to $H - c_i$, and every target is present and retained. Once these folds are done, each of the remaining centres is an apex. At least one remains because $m \geq 2$. If no auxiliary type is present, the same observation gives an apex without any folds. Since folds preserve q and the resulting cone has $q = 0$, we have $q(H - c_i) = 0$.

To identify the other term of the recurrence, take the common neighbours of a and c_i in G_m. These are the neighbours of c_i in H, and also the neighbours of a in the neighbourhood graph of c_i in G_m. The isomorphism of Lemma 3.1 fixes a, so the graph induced by these vertices is isomorphic to $G_{m-1}[N_{G_{m-1}}(a)]$. Thus the recurrence inside H gives

$$q(G_m[N_{G_m}(a)]) = -q(G_{m-1}[N_{G_{m-1}}(a)]).$$

Induction proves (5).

Now fix a type j and a layer i, and put $H = G_m[N_{G_m}(x_{j,i})]$. The vertices of H are those of $A[P_j]$, all copies of the types adjacent to j in R, the copies $x_{j,t}$ with $t \neq i$, and the centres c_t with $t \neq i$. Choose $h \in \mathcal{I}_m$ with $h \neq i$. Both c_h and $x_{j,h}$ belong to H. In this list, $x_{j,h}$ is adjacent to every seed vertex and every other auxiliary, and among the centres it omits only c_h. Hence $x_{j,h}$ is an apex of $H - c_h$, and $q(H - c_h) = 0$.

The graph $H[N_H(c_h)]$ is induced by the common neighbours of $x_{j,i}$ and c_h. In the neighbourhood graph of c_h in G_m, these are the neighbours of $x_{j,i}$. The isomorphism of Lemma 3.1 renumbers layer i as $\rho_h(i)$, so it identifies this graph with $G_{m-1}[N_{G_{m-1}}(x_{j,\rho_h(i)})]$. The recurrence gives

$$q(G_m[N_{G_m}(x_{j,i})]) = -q(G_{m-1}[N_{G_{m-1}}(x_{j,\rho_h(i)})]).$$

At $m-1$ the induction hypothesis applies to the auxiliary of type j in every layer, including the layer with index $\rho_h(i)$. This proves (6). □

In both cases, deleting a suitable centre from the neighbourhood graph leaves a remainder with $q = 0$. The other term of the recurrence is the value on the corresponding neighbourhood one layer

earlier, taken with a minus sign. In the auxiliary case the remainder is already a cone, while for a seed vertex folds may be needed.

3.3. **Strict levels in the layer construction.** The seed determines the level at which we can first delete p. To finish the reduction of B we need a hypothesis on $B - p$, and to exclude first deletions away from p we need the Euler test in B. Section 7 replaces this Euler obstruction by non-zero reduced homology. Lemmas 3.2 and 3.5 carry the corresponding arguments through all subsequent layers.

Theorem 3.6. *Let G_m be the layer construction above, with $B = G_1$ and $p = c_0$. Suppose that, for some integer $s \geq 0$,*

(1) $A \in \mathcal{D}_s \setminus \mathcal{D}_{s-1}$;
(2) $B - p \in \mathcal{D}_s$;
(3) $q(B[N_B(v)]) \neq 0$ *for every* $v \in V(B) \setminus \{p\}$.

Then, for every integer $m \geq 0$,

$$G_m \in \mathcal{D}_{s+m} \setminus \mathcal{D}_{s+m-1}, \qquad |V(G_m)| = |V(A)| + (d+1)m.$$

Proof. We first rule out non-central vertices as candidates for a first deletion. By hypothesis (3) and Lemma 3.5, their neighbourhood graphs in G_m have non-zero q whenever $m \geq 1$. By Lemma 3.4, none of these graphs belongs to a dismantling class. Only centres can start a reduction.

For $m = 0$, hypothesis (1) gives the strict level of the seed. In $G_1 = B$, the neighbourhood graph of p is $A \in \mathcal{D}_s$, so deleting p is admissible at level $s+1$. By hypothesis (2), the remainder $B - p$ belongs to $\mathcal{D}_s \subseteq \mathcal{D}_{s+1}$, so the reduction can be completed at level $s+1$. At level s, the only candidate for a first deletion is p. Its neighbourhood graph is outside $\mathcal{D}_{s-1}$ by hypothesis (1), so the deletion is forbidden. For $s = 0$ this says that the neighbourhood graph is not a non-empty cone. Since B has more than one vertex, Lemma 2.4 gives $B \notin \mathcal{D}_s$.

Now suppose $m \geq 2$ and assume that G_{m-1} has strict level $s + m - 1$. By Lemma 3.1, the neighbourhood graph of any centre of G_m is isomorphic to G_{m-1}. We may therefore delete a centre at level $s + m$ and then use the folds of Lemma 3.2 to complete the reduction, since $\mathcal{D}_0 \subseteq \mathcal{D}_{s+m}$. At level $s + m - 1$, a centre would need its neighbourhood graph in $\mathcal{D}_{s+m-2}$. The inductive hypothesis rules this out, and the Euler obstruction has already excluded the non-central vertices. Thus there is no admissible first deletion, and G_m has more than one vertex. Lemma 2.4 gives $G_m \notin \mathcal{D}_{s+m-1}$. The order formula is (2). □

Remark 3.7. *The proof uses the remainder hypothesis only to obtain $B - p \in \mathcal{D}_{s+1}$ in the first step. Thus the conclusion also holds if hypothesis (2) is replaced by this weaker condition, while hypotheses (1) and (3) are unchanged.*

Each layer contributes a centre and d auxiliaries. The Euler criterion itself puts a lower bound on d. If the seed and the remainder both belong to dismantling classes, the neighbourhood graph of a sole auxiliary has $q = 0$.

Proposition 3.8. *Any data satisfying the hypotheses of Theorem 3.6 have $d \geq 2$. Thus this construction with its Euler obstruction adds at least three vertices per layer. The same conclusion holds under the weaker remainder hypothesis of Remark 3.7.*

Proof. The definition takes $d \geq 1$. Even if we allowed $d = 0$, the first extension would consist of A and its new universal centre p. Since A is non-empty, choose $a \in V(A)$. Then p is an apex of $B[N_B(a)]$, including when a has no neighbours in A. By Lemma 3.3, $q(B[N_B(a)]) = 0$, in contradiction to hypothesis (3).

For $d = 1$, the sole auxiliary x_0 has neighbourhood graph $A[P_0]$ both in B and in $B - p$. Deleting x_0 from $B - p$ leaves A. Applying Lemma 3.3 to this deletion gives

$$q(B - p) = q(A) - q(A[P_0]).$$

By hypotheses (1) and (2), both A and $B - p$ belong to dismantling classes, so $q(A) = q(B - p) = 0$ by Lemma 3.4. The displayed identity then gives $q(B[N_B(x_0)]) = q(A[P_0]) = 0$, in contradiction to hypothesis (3). The same calculation applies when $B - p \in \mathcal{D}_{s+1}$, since q vanishes on that class as well. □

The bound applies to the layer construction under its Euler hypotheses. The family F_k of the article has two auxiliary types and attains this cost of three vertices per layer. Appendix A records its finite data; the next two sections give the other two applications.

4. The family of order $4k + 4$

The second family in the article begins with graphs on eight, twelve and sixteen vertices at strict levels one, two and three. All three occur as nested induced subgraphs. From the sixteen-vertex seed onwards, the layer construction adds four vertices at a time. We specify these graphs, justify their finite reductions and obstructions, and then apply the general theorem.

4.1. **The sixteen-vertex seed and its profiles.** We use three auxiliary types, so each layer adds four vertices. Let $V(A) = \{a_0, \ldots, a_{15}\}$, with the adjacency list in Table 1. It is symmetric, has no loops, and has 97 edges. The profiles in Table 2 have sizes 13, 14 and 15. The type graph is

$$E(R) = \{\{0, 1\}, \{0, 2\}\}. \tag{7}$$

TABLE 1. Complete adjacency list of the sixteen-vertex seed A.

v	$N_A(a_v)$, given by indices
0	2, 3, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14
1	2, 3, 4, 6, 7, 8, 9, 10, 11, 12, 13, 14
2	0, 1, 3, 6, 7, 8, 9, 10, 11, 12, 13, 15
3	0, 1, 2, 4, 5, 8, 9, 10, 11, 12, 13, 15
4	1, 3, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15
5	0, 3, 4, 6, 7, 9, 10, 11, 12, 13, 14, 15
6	0, 1, 2, 4, 5, 8, 9, 11, 12, 13, 14, 15
7	0, 1, 2, 4, 5, 8, 9, 10, 12, 13, 14, 15
8	0, 1, 2, 3, 4, 6, 7, 9, 10, 12, 14, 15
9	0, 1, 2, 3, 5, 6, 7, 8, 11, 13, 14, 15
10	0, 1, 2, 3, 4, 5, 7, 8, 11, 13, 14, 15
11	0, 1, 2, 3, 4, 5, 6, 9, 10, 12, 14, 15
12	0, 1, 2, 3, 4, 5, 6, 7, 8, 11, 14, 15
13	0, 1, 2, 3, 4, 5, 6, 7, 9, 10, 14, 15
14	0, 1, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 15
15	2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14

TABLE 2. Profiles specified by their omitted base vertices. Every copy of type j is adjacent to all remaining vertices of A.

j	$V(A) \setminus P_j$	$\lvert P_j\rvert$
0	$\{a_8, a_{11}, a_{14}\}$	13
1	$\{a_{11}, a_{15}\}$	14
2	$\{a_{13}\}$	15

The graph $B = G_1$ adds $p = c_0$ and x_0, x_1, x_2 to A. The centre is adjacent to every vertex of A and to no auxiliary. The neighbours of x_j in A are P_j, and its auxiliary adjacencies are specified by R. Thus

$$|V(B)| = 20, \qquad |E(B)| = 97 + 16 + 42 + 2 = 157.$$

For the finite certificates, a_i has index i, $p = 16$, and $x_j = 17 + j$.

4.2. **The eight- and twelve-vertex prefixes.** The first two members are induced subgraphs of the same seed. Put

$$S_1 = \{2, 4, 5, 8, 9, 10, 12, 13\},$$
$$S_2 = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13\},$$

and define

$$H_1 = A[\{a_i : i \in S_1\}], \qquad H_2 = A[\{a_i : i \in S_2\}], \qquad H_3 = A.$$

The direct reduction below establishes the strict level of the first prefix. The recursive certificates then establish the strict level of the second prefix and of the sixteen-vertex seed. The two prefixes have 8 and 12 vertices, respectively. The inclusions $S_1 \subset S_2 \subset V(A)$ give induced inclusions. We

now verify the first prefix directly. Use numerical labels $a_i = i$. Restricting the adjacency table of A gives

v	$N_{H_1}(v)$
2	8, 9, 10, 12, 13
4	5, 8, 10, 12, 13
5	4, 9, 10, 12, 13
8	2, 4, 9, 10, 12
9	2, 5, 8, 13
10	2, 4, 5, 8, 13
12	2, 4, 5, 8
13	2, 4, 5, 9, 10

Proposition 4.1. *The graph H_1 belongs to $\mathcal{D}_1 \setminus \mathcal{D}_0$.*

Proof. First consider $H_1[N_{H_1}(5)]$, on $\{4, 9, 10, 12, 13\}$. The only neighbour of 9 there is 13, so $9 \to 13$ is a fold. The remaining graph on $\{4, 10, 12, 13\}$ is a cone with apex 4. Consequently 5 can be deleted from H_1 at level one.

In $H_1 - 5$, the neighbourhood of 4 is $\{8, 10, 12, 13\}$. Within this neighbourhood, $12 \to 8$ is a fold, and the remaining path 8, 10, 13 is a cone with apex 10. Thus 4 can next be deleted at level one. The graph $H_1 - \{5, 4\}$ is a cone with apex 2, proving $H_1 \in \mathcal{D}_1$. The inner folds justify only the stated outer deletion; they do not delete vertices from the outer graph.

To exclude level zero, Table 3 lists, for every v, each candidate apex $t \in N_{H_1}(v)$ and a vertex $w \in N_{H_1}(v) \setminus \{t\}$ not adjacent to it. Every candidate occurs, so no neighbourhood has an apex. There is no first fold, and $H_1 \notin \mathcal{D}_0$. □

TABLE 3. Complete exclusion of an apex in every neighbourhood of H_1.

v	Candidate apex $t \mapsto w$, with $tw \notin E(H_1)$
2	$8 \mapsto 13,\ 9 \mapsto 10,\ 10 \mapsto 9,\ 12 \mapsto 9,\ 13 \mapsto 8$
4	$5 \mapsto 8,\ 8 \mapsto 5,\ 10 \mapsto 12,\ 12 \mapsto 10,\ 13 \mapsto 8$
5	$4 \mapsto 9,\ 9 \mapsto 4,\ 10 \mapsto 9,\ 12 \mapsto 9,\ 13 \mapsto 12$
8	$2 \mapsto 4,\ 4 \mapsto 2,\ 9 \mapsto 4,\ 10 \mapsto 9,\ 12 \mapsto 9$
9	$2 \mapsto 5,\ 5 \mapsto 2,\ 8 \mapsto 5,\ 13 \mapsto 8$
10	$2 \mapsto 4,\ 4 \mapsto 2,\ 5 \mapsto 2,\ 8 \mapsto 5,\ 13 \mapsto 8$
12	$2 \mapsto 4,\ 4 \mapsto 2,\ 5 \mapsto 2,\ 8 \mapsto 5$
13	$2 \mapsto 4,\ 4 \mapsto 2,\ 5 \mapsto 2,\ 9 \mapsto 4,\ 10 \mapsto 9$

4.3. **Certifying the prefixes and the seed.** For the larger graphs, the neighbourhood reductions are themselves recursive. We record them as finite proof states so that every required membership and non-membership can be checked. The main certificate contains 1,114 proof nodes for A and 184 for $B - p$. The prefix certificates use the ordered vertex lists S_1, S_2 of this section. All neighbourhoods and domination relations are computed in the current induced subgraph.

4.3.1. *The recursive proof rule.* A proof state (S, j), with $j \geq -1$, asserts whether the graph induced by S belongs to $\mathcal{D}_j$. There are five allowed forms of node.

(1) The empty state is false at every level.

(2) A non-empty cone is true at every level, with its apex supplied.
(3) A non-empty non-cone is false at level -1.
(4) A true state with $j \geq 0$ specifies $v \in S$, a true child $(S \cap N(v), j-1)$, and a true child $(S \setminus \{v\}, j)$.
(5) A non-empty non-cone state (S, j), with $j \geq 0$ (and hence $|S| \geq 2$), is false if it specifies, for every $v \in S$, a false child of at least one of those two forms.

The checker verifies the exact subsets, levels, truth values and child references, requires every vertex in a negative node to be covered, and rejects cycles. Every child has fewer vertices. It reconstructs each induced graph from the stated adjacency data, tests every proposed apex, and checks the absence of an apex in each negative non-cone state. A deleted vertex must belong to the current set. The correctness of a checked proof graph therefore follows by induction on the number of vertices, using Definition 2.2 and Lemma 2.3 for cone states. By the first-deletion equivalence (1), any reduction of a state with $|S| \geq 2$ begins with a vertex v such that $(S \cap N(v), j-1)$ is true, and the remaining reduction shows that $(S \setminus \{v\}, j)$ is true. The false children of a form-5 node exclude every such v. This gives a finite certificate of non-membership as well as membership, without relying on a bounded unsuccessful search.

The three main roots are bound to

$$(V(A), 3, \text{true}), \quad (V(A), 2, \text{false}), \quad (V(B) \setminus \{p\}, 3, \text{true}).$$

The four prefix roots assert membership at levels $1, 2$ and non-membership at levels $0, 1$, respectively. The checker also verifies the prefix adjacencies against the specified induced subgraphs of A. The seed and first extension are reconstructed from Tables 1 and 2, the type graph in (7), and the six adjacency rules of Section 3.

The complete finite proof graphs are the files `data/certificates/FINITE_CERTIFICATE.json.gz` and `data/certificates/PREFIX_CERTIFICATES.json.gz`. The seed and profiles are in `data/certificates/PARAMETERS.json`; the ordered prefix sets and their adjacencies are in `data/certificates/PREFIX.json`. The checking rules are implemented in `dismantling/certificates.py`. The verifier checks the supplied proof states using integer and set operations; it does not search for reductions.

4.3.2. *The reductions and their obstructions.* For H_1, delete a_5, a_4 in that order; the remaining graph is a cone with apex a_2. Their current neighbourhoods belong to $\mathcal{D}_0$, as proved by the referenced child certificates. For H_2, delete a_6, a_5, a_4; the current neighbourhoods belong to $\mathcal{D}_1$, and the remainder is a cone with apex a_2. For $A = H_3$, delete a_0, a_1; the current neighbourhoods belong to $\mathcal{D}_2$, and the remainder is a cone with apex a_{15}. For $B - p$, the outer order is

$$a_3, a_1, a_2, a_0, a_4, a_6, a_5, a_8, a_9, a_{10}, a_{11}.$$

Each current neighbourhood belongs to $\mathcal{D}_2$, and the remainder is a cone with apex a_7. These orders alone do not certify their neighbourhoods; the complete child proof graphs do so by the preceding rules.

The negative root for $A \notin \mathcal{D}_2$ has one obstruction for each of its 16 vertices: every $A[N_A(a_i)]$ is certified outside $\mathcal{D}_1$. The prefix negative roots similarly provide a complete obstruction for every possible first deletion at their claimed lower levels. A zero value of q is never used as a sufficient test of membership.

4.4. **The nineteen neighbourhood obstructions.** Table 4 supplies all coefficients through degree seven of every non-central neighbourhood in B. All higher degrees were computed and verified to vanish. They are checked using the coefficient recurrence (4). The supplied checker uses this deletion recurrence to verify every coefficient and alternating sum in the distributed certificate. For example, the row for a_0 gives $1 - 16 + 95 - 263 + 344 - 182 + 20 = -1$. All 19 sums are non-zero.

TABLE 4. Clique coefficients of $B[N_B(v)]$, for every non-central vertex. All coefficients of degree greater than seven are zero. The last column is their alternating sum.

v	c_0	c_1	c_2	c_3	c_4	c_5	c_6	c_7	q
a_0	1	16	95	263	344	182	20	0	−1
a_1	1	16	95	262	340	179	20	0	−1
a_2	1	16	96	271	362	189	10	0	−7
a_3	1	16	95	261	335	172	16	0	−2
a_4	1	16	96	275	391	257	62	3	−1
a_5	1	16	97	281	402	262	60	3	−2
a_6	1	16	95	263	349	200	34	2	−2
a_7	1	16	97	279	388	232	39	1	−3
a_8	1	15	86	235	308	164	15	0	−4
a_9	1	16	95	263	344	183	19	0	−3
a_{10}	1	16	95	264	354	207	37	1	−1
a_{11}	1	14	75	194	249	143	27	2	−1
a_{12}	1	16	96	271	367	211	34	2	−2
a_{13}	1	15	86	237	323	199	41	1	−1
a_{14}	1	16	97	281	402	262	60	3	−2
a_{15}	1	16	96	272	375	230	48	3	−1
x_0	1	15	86	236	315	179	26	0	−2
x_1	1	15	85	228	296	167	27	0	−1
x_2	1	16	97	278	380	213	27	1	−3

Reproduce this calculation. From the repository directory, run

```
python reproduce.py -only 4k4
```

The output reports seven checked roots and nineteen clique rows. The root checks certify $H_1 \in \mathcal{D}_1 \setminus \mathcal{D}_0$, $H_2 \in \mathcal{D}_2 \setminus \mathcal{D}_1$, $A \in \mathcal{D}_3 \setminus \mathcal{D}_2$, and $B - p \in \mathcal{D}_3$. Every coefficient in the printed table is compared with both clique enumeration and the deletion recurrence, including the vanishing of higher coefficients.

4.5. **The family at every level.** The certificates establish the hypotheses needed for the extension from the sixteen-vertex seed. Theorem 3.6 then treats any number of layers.

Theorem 4.2. *With the seed and profiles in this section, put* $H_3 = A$ *and* $H_k = G_{k-3}$ *for* $k \geq 3$*, and use the two induced prefixes for* $k = 1, 2$*. Then*

$$H_k \in \mathcal{D}_k \setminus \mathcal{D}_{k-1}, \qquad |V(H_k)| = 4k + 4 \qquad (k \geq 1).$$

Every inclusion $H_k \subset H_{k+1}$ *is induced.*

Proof. The seven roots give the two prefixes and establish $A \in \mathcal{D}_3 \setminus \mathcal{D}_2$ and $B - p \in \mathcal{D}_3$. The nineteen non-zero sums in Table 4 exclude every non-central vertex as a first deletion. Thus Theorem 3.6 applies with $s = 3$ and $d = 3$. For $k \geq 3$ it gives strict level k and $16 + 4(k-3) = 4k + 4$ vertices. The prefix certificates prove the cases $k = 1, 2$. The nested prefix sets and the layer rules give the induced inclusions. □

For $k \geq 3$, put $m = k - 3$. The layered graphs also have

$$|E(G_m)| = 97 + 16m + 42m + 4\binom{m}{2} + 2m^2 + 3m(m-1) = 97 + 53m + 7m^2.$$

The terms count seed edges, seed–centre edges, seed–auxiliary edges, edges between centres or within one type, edges between the two adjacent pairs of types, and centre–auxiliary edges, respectively.

5. The family of order $6k + 6$

The same layer construction also applies to a twelve-vertex seed at strict level one. This example uses five auxiliary types, and its positive certificates can be written out entirely as folds inside successive neighbourhoods. It illustrates how the three finite hypotheses are checked when no recursive certificate file is needed. We first state the finite assertions and their consequences; the reductions and clique tables used in the proof are displayed in Section 5.2.

5.1. **The seed and the resulting family.** Here five auxiliary types give six new vertices per layer. Let $V(A) = \{a_0, \ldots, a_{11}\}$, with the adjacency list in Table 5. The list is symmetric, has no loops, and gives $|E(A)| = 47$. The profiles are given in Table 6; their cardinalities sum to 48. The type graph is

$$E(R) = \{02, 03, 04, 12, 14, 23, 34\}, \tag{8}$$

where jt abbreviates the unordered pair $\{j, t\}$. There are no other edges of R.

TABLE 5. The complete adjacency list of A. An integer i denotes a_i.

v	$N_A(v)$
0	$\{1, 3, 4, 5, 6, 7, 9, 10\}$
1	$\{0, 2, 3, 4, 5, 6, 7, 8, 10\}$
2	$\{1, 4, 5, 6, 8, 9, 11\}$
3	$\{0, 1, 4, 6, 7, 8, 9, 11\}$
4	$\{0, 1, 2, 3, 5, 8, 10, 11\}$
5	$\{0, 1, 2, 4, 6, 9, 10, 11\}$
6	$\{0, 1, 2, 3, 5, 8, 9\}$
7	$\{0, 1, 3, 9, 10, 11\}$
8	$\{1, 2, 3, 4, 6, 9, 10, 11\}$
9	$\{0, 2, 3, 5, 6, 7, 8, 10, 11\}$
10	$\{0, 1, 4, 5, 7, 8, 9, 11\}$
11	$\{2, 3, 4, 5, 7, 8, 9, 10\}$

TABLE 6. The five profiles. Each displayed integer i denotes a_i.

j	Indices of the vertices in P_j
0	$1,2,3,4,5,6,7,8,9,11$
1	$1,3,4,6,7,8,10,11$
2	$0,1,3,4,5,6,7,9,10,11$
3	$1,2,3,5,6,7,8,9,10,11$
4	$0,1,2,3,5,7,8,9,10,11$

The graph B consists of A, the centre p, and the five auxiliaries $x_0,\dots,x_4$. Its centre is adjacent to every vertex of A and to no auxiliary; the neighbours of x_j in A are precisely P_j, and its auxiliary adjacencies are those of R. Thus

$$|V(B)| = 18, \qquad |E(B)| = 47 + 12 + 48 + 7 = 114.$$

In the finite certificates, a_i is abbreviated to i, and $p = 12$, $x_j = 13 + j$. In particular, vertex 15 denotes x_2.

Proposition 5.1. *These seed data satisfy*

$$A \in \mathcal{D}_1 \setminus \mathcal{D}_0, \qquad B - p \in \mathcal{D}_1, \qquad q(B[N_B(v)]) = \begin{cases} 2, & v = a_7 \text{ or } v = x_2, \\ 1, & v \neq p, a_7, x_2 \end{cases} \quad (v \neq p).$$

Proof. Table 7 gives, for every $v \in V(A)$, the degree $d_v = |N_A(v)|$ and the maximum degree in $A[N_A(v)]$. In each row that maximum is strictly smaller than $d_v - 1$. No open neighbourhood graph is a cone; Lemma 2.4 therefore gives $A \notin \mathcal{D}_0$.

Tables 8 and 9 specify successive outer deletions in A and $B - p$, respectively. Each row lists the exact neighbourhood set in the current outer graph, the folds within that neighbourhood in the stated order, and an apex of the resulting neighbourhood. All fourteen required closed neighbourhood inclusions are displayed in Table 10. They are obtained by restricting the adjacency data of Tables 5 and 6 and (8) to the current set. After the outer deletions, the remaining vertex sets are cones with the apices specified in Section 5.2. Each inner reduction is consequently a level-0 reduction, and each outer deletion is permitted at level 1. The final cones also lie in $\mathcal{D}_1$, by Lemma 2.3. This proves $A \in \mathcal{D}_1$ and $B - p \in \mathcal{D}_1$.

Table 11 gives all clique coefficients of the seventeen non-central neighbourhood graphs. These coefficients follow by the coefficient recurrence in Lemma 3.3, with $c_0(H) = 1$ and $c_t(\varnothing) = 0$ for $t \geq 1$. Their alternating sums are 2 in rows 7 and 15, and 1 in every other row. This proves the last assertion. □

For this seed, the graphs of Section 3 satisfy

$$|V(G_m)| = 12 + 6m, \qquad |E(G_m)| = 47 + 52m + 15m^2. \tag{9}$$

Indeed, the seven disjoint classes of edges give

$$\begin{aligned} |E(G_m)| &= 47 + 12m + 48m + \binom{m}{2} + 5\binom{m}{2} + 7m^2 + 5m(m-1) \\ &= 47 + 52m + 15m^2. \end{aligned}$$

The terms count, in order, edges within A, between A and centres, between A and auxiliaries, between centres, within each auxiliary type, between distinct types, and between centres and auxiliaries.

Theorem 5.2. *For every* $k \geq 1$, *put* $J_k = G_{k-1}$. *Then* $J_k \in \mathcal{D}_k \setminus \mathcal{D}_{k-1}$ *and* $|V(J_k)| = 6k + 6$.

Proof. Proposition 5.1 supplies the three hypotheses of Theorem 3.6, with $s = 1$ and $d = 5$. Consequently

$$G_m \in \mathcal{D}_{m+1} \setminus \mathcal{D}_m \qquad (m \geq 0).$$

Taking $m = k - 1$ gives $J_k \in \mathcal{D}_k \setminus \mathcal{D}_{k-1}$ and $|V(J_k)| = 12 + 6(k-1) = 6k + 6$. □

5.2. **The finite reductions and obstructions.** The certificates below use only the adjacency list, the profiles, and the type edges specified in Section 5.1.

5.2.1. *Excluding a first fold in* A. For $v \in V(A)$, put $d_v = |N_A(v)|$, and write $\Delta(Q)$ for the maximum degree in a non-empty graph Q. The entries of Table 7 satisfy $\Delta(A[N_A(v)]) < d_v - 1$, so none of these neighbourhoods has an apex.

TABLE 7. Every open neighbourhood graph of A has no universal vertex.

v	$d_v = \lvert N_A(v)\rvert$	$d_v - 1$	$\Delta(A[N_A(v)])$
0	8	7	6
1	9	8	6
2	7	6	5
3	8	7	5
4	8	7	6
5	8	7	5
6	7	6	5
7	6	5	4
8	8	7	5
9	9	8	6
10	8	7	5
11	8	7	6

5.2.2. *The two level-1 reductions.* In each row of Tables 8 and 9, v is deleted from the outer graph after all preceding outer deletions. The set S is its neighbourhood set at that moment. The arrows are folds inside the graph induced by S. After each arrow its source vertex is removed before the next inclusion is evaluated. The listed apex is universal after those inner folds. A row with no folds already has a conical neighbourhood graph.

For Table 8, the final outer vertex set is $\{5, 6, 8, 9, 10, 11\}$, with apex 9. For Table 9, it is $\{3, 7, 8, 9, 10, 11, 13, 14, 15, 16, 17\}$, with apex 11. The centre p has been removed before Table 9 begins.

TABLE 8. A reduction certificate for $A \in \mathcal{D}_1$.

v	Current neighbourhood set S	Folds in order	Apex
7	$\{0,1,3,9,10,11\}$	$1 \to 0$	9
0	$\{1,3,4,5,6,9,10\}$	$4 \to 1$, $3 \to 6$	5
1	$\{2,3,4,5,6,8,10\}$	$3 \to 8$, $6 \to 2$	4
2	$\{4,5,6,8,9,11\}$	$4 \to 11$	9
3	$\{4,6,8,9,11\}$	none	8
4	$\{5,8,10,11\}$	none	10

TABLE 9. A reduction certificate for $B - p \in \mathcal{D}_1$.

v	Current neighbourhood set S	Folds in order	Apex
0	$\{1,3,4,5,6,7,9,10,15,17\}$	$4 \to 1$, $6 \to 15$, $3 \to 7$	10
2	$\{1,4,5,6,8,9,11,13,16,17\}$	none	13
4	$\{1,3,5,8,10,11,13,14,15\}$	$5 \to 15$, $10 \to 14$	3
1	$\{3,5,6,7,8,10,13,14,15,16,17\}$	$5 \to 16$, $6 \to 3$, $8 \to 17$	7
5	$\{6,9,10,11,13,15,16,17\}$	none	9
6	$\{3,8,9,13,14,15,16\}$	none	3

For example, the first row of Table 8 starts with $S = \{0,1,3,9,10,11\}$. Within this neighbourhood,

$$N_{A[S]}[1] = \{0,1,3,10\} \subseteq N_{A[S]}[0] = \{0,1,3,9,10\}.$$

The fold $1 \to 0$ leaves a cone with apex 9, so the outer vertex 7 can be deleted at level 1. The next row is evaluated in $A - 7$. Table 10 provides the corresponding sets for every fold in both certificates. Its third column is contained in its fourth in each row; both columns are the closed neighbourhoods in the current inner graph.

TABLE 10. Closed neighbourhood inclusions for all fourteen folds. Here T is the neighbourhood graph after the preceding folds in the same row of the corresponding certificate.

Graph; outer v	Fold $u \to t$	$N_T[u]$	$N_T[t]$
A; 7	$1 \to 0$	$\{0,1,3,10\}$	$\{0,1,3,9,10\}$
A; 0	$4 \to 1$	$\{1,3,4,5,10\}$	$\{1,3,4,5,6,10\}$
A; 0	$3 \to 6$	$\{1,3,6,9\}$	$\{1,3,5,6,9\}$
A; 1	$3 \to 8$	$\{3,4,6,8\}$	$\{2,3,4,6,8,10\}$
A; 1	$6 \to 2$	$\{2,5,6,8\}$	$\{2,4,5,6,8\}$
A; 2	$4 \to 11$	$\{4,5,8,11\}$	$\{4,5,8,9,11\}$
B − p; 0	$4 \to 1$	$\{1,3,4,5,10,15\}$	$\{1,3,4,5,6,7,10,15,17\}$
B − p; 0	$6 \to 15$	$\{1,3,5,6,9,15\}$	$\{1,3,5,6,7,9,10,15\}$
B − p; 0	$3 \to 7$	$\{1,3,7,9,15,17\}$	$\{1,3,7,9,10,15,17\}$
B − p; 4	$5 \to 15$	$\{1,5,10,11,13,15\}$	$\{1,3,5,10,11,13,14,15\}$
B − p; 4	$10 \to 14$	$\{1,8,10,11,14,15\}$	$\{1,3,8,10,11,14,15\}$
B − p; 1	$5 \to 16$	$\{5,6,10,13,15,16,17\}$	$\{3,5,6,7,8,10,13,15,16,17\}$
B − p; 1	$6 \to 3$	$\{3,6,8,13,14,15,16\}$	$\{3,6,7,8,13,14,15,16,17\}$
B − p; 1	$8 \to 17$	$\{3,8,10,13,14,16,17\}$	$\{3,7,8,10,13,14,16,17\}$

5.2.3. *Clique coefficients.* Table 11 supplies the coefficients $c_0, \ldots, c_6$ of every non-central neighbourhood graph in B. There are no cliques of greater cardinality in these graphs. The coefficient recurrence (4) computes every row from the adjacency data. For instance, the alternating sums in rows 0 and 7 are

$$1 - 11 + 41 - 63 + 35 - 2 = 1, \qquad 1 - 12 + 49 - 86 + 63 - 15 + 2 = 2.$$

All seventeen sums are non-zero.

TABLE 11. Clique counts for the seventeen non-central open neighbourhood graphs of B. Coefficients c_t with $t > 6$ are zero.

v	c_0	c_1	c_2	c_3	c_4	c_5	c_6	q
0	1	11	41	63	35	2	0	1
1	1	15	72	133	88	12	0	1
2	1	11	43	75	58	17	2	1
3	1	14	65	122	91	23	3	1
4	1	12	46	71	39	2	0	1
5	1	13	57	103	74	16	1	1
6	1	12	46	75	51	10	0	1
7	1	12	49	86	63	15	2	2
8	1	13	57	103	75	18	2	1
9	1	14	66	131	111	37	5	1
10	1	13	56	97	61	7	0	1
11	1	14	65	125	101	32	5	1
13	1	13	59	119	110	42	5	1
14	1	10	33	45	22	0	0	1
15	1	13	57	103	74	15	1	2
16	1	13	60	125	120	47	5	1
17	1	13	56	104	85	28	4	1

Reproduce this calculation. From the repository directory, run

```
python reproduce.py -only 6k6
```

The output confirms twelve outer deletions, fourteen inner folds, twelve rows excluding a first fold in the seed, and seventeen clique rows. Each displayed neighbourhood and both sides of every fold inclusion are recomputed from the adjacency lists. The calculation also verifies the final cones in the two reductions.

6. COMPLETING KERNELS WITH TWO AUXILIARY TYPES

The family F_k adds only two auxiliaries per layer. For this case we can understand the neighbourhood obstructions by removing certain edges from the seed. We first introduce the homology used to compare the resulting graphs, then derive the identities and apply them to the eleven-vertex seed.

6.1. **Homology for the comparison.**

A non-zero reduced homology group excludes a graph from every dismantling class. We give the chain conventions and the elementary facts used in this section and in Section 7; in particular, empty neighbourhoods must remain visible in degree -1. The clique complex $\mathrm{Cl}(H)$ contains the cliques of H, including the empty face. We use *augmented reduced homology over* $\mathbb{F}_2$. For a complex K, the chain group $\widetilde{C}_t(K)$ has the faces of cardinality $t+1$ as a basis when $t \geq 0$, and $\widetilde{C}_{-1}(K) = \mathbb{F}_2$, generated by the empty face. Lower chain groups are zero. The boundary of a face is the sum of its codimension-one faces; in particular the boundary of a vertex is the empty face. Each term occurs twice in ∂^2, so this is a chain complex.

We write $\widetilde{H}_t(K) = \ker \partial_t / \operatorname{im} \partial_{t+1}$ and call K *acyclic* if every $\widetilde{H}_t(K)$ vanishes. In particular,

$$\widetilde{H}_{-1}(\{\varnothing\}) = \mathbb{F}_2, \qquad \widetilde{H}_t(\{\varnothing\}) = 0 \quad (t \neq -1). \tag{10}$$

A complex with a vertex has zero homology in degree -1. The *reduced Betti number* of a complex K in degree t is $\widetilde{\beta}_t(K) = \dim_{\mathbb{F}_2} \widetilde{H}_t(K)$. Acyclicity is used as a homological condition, not as a synonym for contractibility or for membership in any $\mathcal{D}_j$. When a homological term is applied to a graph, it refers to its clique complex; in particular, *acyclic* here does not mean that the graph is a forest.

A *simplicial cone* with apex w is a complex $w * J$ whose faces are exactly the faces σ of J and the sets $\sigma \cup \{w\}$, where $w \notin V(J)$. In particular, if a graph H is a cone with apex w, then $\mathrm{Cl}(H) = w * \mathrm{Cl}(H - w)$.

Lemma 6.1. *A simplicial cone is acyclic.*

Proof. For an apex w, let h adjoin w to a face not containing it and send a face containing w to zero. Direct expansion gives $\partial h + h\partial = \mathrm{id}$, including on the empty face. Every cycle z is therefore the boundary of $h(z)$. □

We will compare complexes by expressing them as unions of acyclic pieces.

Lemma 6.2. *If $M = M_1 \cup M_2$ and M_1, M_2 are acyclic, then for every integer t,*

$$\widetilde{H}_t(M) \cong \widetilde{H}_{t-1}(M_1 \cap M_2).$$

The intersection may be $\{\varnothing\}$.

Proof. For $t \geq 0$, write a cycle $z = u + v$, with chains in M_1, M_2. Then $\partial u = \partial v = b$ is a cycle in $M_1 \cap M_2$. Changing the decomposition adds the same chain of the intersection to u and v, so it changes b by a boundary. If z is changed by a boundary, decompose a filling into chains of M_1, M_2; the resulting class of b is unchanged. This defines $[z] \mapsto [b]$.

Every cycle b of the intersection has fillings u in M_1 and v in M_2, since both complexes are acyclic; then $u + v$ proves surjectivity. If $b = \partial a$ in the intersection, the chains $u + a, v + a$ are cycles in M_1, M_2 and hence boundaries. Their sum is z, proving injectivity. For $t = -1$, the left side is zero since the acyclic complexes M_1, M_2 have vertices; the right side is in degree -2. In lower degrees both sides vanish. □

The folds used in the graph reductions preserve these homology groups.

Lemma 6.3. *A fold $v \to w$ induces isomorphisms $\widetilde{H}_t(\mathrm{Cl}(H - v)) \cong \widetilde{H}_t(\mathrm{Cl}(H))$ for every integer t.*

Proof. Pair every face σ containing v but not w with $\tau = \sigma \cup \{w\}$. The domination condition guarantees that τ is a face. Remove these pairs in decreasing order of $|\sigma|$. When a pair is removed, τ is the only proper coface of σ still present: every larger coface was removed in an earlier pair. The remaining faces form a subcomplex.

To see that removing one such pair preserves augmented homology, let K' be the resulting subcomplex of the current complex K and put $r = \dim \sigma$. A cycle of degree $r + 1$ has coefficient zero at τ, since its boundary has coefficient zero at σ. For a cycle of degree r, adding its coefficient at σ times $\partial\tau$ removes σ without changing its homology class. In other degrees no basis element is removed. Thus every class has a representative in K'.

Conversely, suppose a cycle z' in K' bounds a chain y in K. If y has degree $r + 1$, its coefficient at τ is zero, because the coefficient of σ in z' is zero. If y has degree r, add its coefficient at σ times $\partial\tau$;

this removes σ and preserves its boundary. The resulting filling lies in K'. This proves injectivity as well, including in degree -1. After all pairs are removed the complex is exactly $\mathrm{Cl}(H - v)$. □

The recursive neighbourhood condition therefore gives a homological obstruction at every level.

Lemma 6.4. *If $H \in \mathcal{D}_j$, $j \geq -1$, then $\mathrm{Cl}(H)$ is acyclic.*

Proof. Cones are covered by Lemma 6.1. Induct on j and, at a fixed level, on the number of vertices. The one-vertex graph is a cone. For a valid first deletion v, the two induction hypotheses give acyclicity of $\mathrm{Cl}(H - v)$ and $\mathrm{Cl}(L_H(v))$. The exact decomposition

$$\mathrm{Cl}(H) = \mathrm{Cl}(H - v) \cup \big(v * \mathrm{Cl}(L_H(v))\big)$$

has acyclic pieces and intersection $\mathrm{Cl}(L_H(v))$. Lemma 6.2 proves the assertion. □

6.2. **The extension and its kernel.** In a construction with two auxiliary types joined by an edge, the profiles determine which seed edges contribute to the homological obstructions. Let A be a non-empty graph, and choose $S, T \subseteq V(A)$. Form B by adding vertices p, x, y. The centre p is adjacent to all of A and to neither auxiliary vertex. The neighbours of x in A are exactly S, those of y are exactly T, and xy is an edge. The induced graph on A remains unchanged.

Lemma 6.5. *If every non-central neighbourhood of B has non-trivial reduced homology, then $S \cup T = V(A)$. The same conclusion holds if $q(L_B(v)) \neq 0$ for every $v \neq p$.*

Proof. If $a \notin S \cup T$, then its neighbours in B are $N_A(a)$ and p. The centre is an apex of $L_B(a)$. This cone has zero reduced homology and $q = 0$, contradicting either condition. □

Henceforth assume $S \cup T = V(A)$, and put

$$S_0 = S \setminus T, \qquad T_0 = T \setminus S, \qquad Z = S \cap T.$$

Define H by deleting from A all its edges between S_0 and T_0, and no others. We call H the *kernel of these profile data*. This name describes an operation on edges; it does not impose a homological property on H.

Set

$$K = \mathrm{Cl}(A), \quad X = \mathrm{Cl}(A[S]), \quad Y = \mathrm{Cl}(A[T]), \quad W = \mathrm{Cl}(A[Z]), \quad U = \mathrm{Cl}(H).$$

Every clique of H is contained in S or in T, since it cannot meet both S_0 and T_0. Conversely every clique contained in S or in T survives the edge deletions. Therefore

$$U = X \cup Y, \qquad W = X \cap Y. \tag{11}$$

In particular this union is a clique complex, even if it is smaller than K.

6.3. **The two auxiliary stars.** To compare the seed, its extension and the remainder, we group the faces containing auxiliary vertices. For a vertex z outside a complex J, $z * J$ denotes its cone. Define the subcomplex

$$V = (x * X) \cup \big(y * (Y \cup (x * W))\big). \tag{12}$$

It consists of all faces using an auxiliary vertex together with their subfaces. Its two displayed pieces are cones and their intersection is $x * W$, also a cone. This intersection contains the vertex x,

even when $W = \{\varnothing\}$. By Lemma 6.2, V is acyclic. The faces of V on the original vertices are exactly $X \cup Y$. Consequently

$$\mathrm{Cl}(B) = (p * K) \cup V, \qquad \mathrm{Cl}(B - p) = K \cup V, \qquad (p * K) \cap V = K \cap V = U. \tag{13}$$

These are equalities of complexes.

Proposition 6.6. *For every integer* t,

$$\widetilde{H}_t(\mathrm{Cl}(B)) \cong \widetilde{H}_{t-1}(U).$$

If K *is acyclic, then also*

$$\widetilde{H}_t(\mathrm{Cl}(B - p)) \cong \widetilde{H}_{t-1}(U).$$

Under this extra hypothesis, $\mathrm{Cl}(B - p)$ *is acyclic if and only if* U *is acyclic.*

Proof. The two pieces in the first decomposition of (13) are acyclic and intersect in U. Apply Lemma 6.2. For the second decomposition the same argument applies when K is acyclic. The shift covers every possible non-zero degree; hence the last equivalence follows. □

For a seed $A \in \mathcal{D}_s$, membership of $B - p$ in a dismantling class therefore implies acyclicity of the kernel. The converse statement about membership does not follow: the proposition characterises acyclicity of the remainder, not its dismantling level.

6.4. **Original and auxiliary neighbourhoods.** The global decomposition also applies within each neighbourhood. For an original vertex, its intersection is the neighbourhood complex in the kernel.

Theorem 6.7. *For every* $a \in V(A)$ *and every integer* t,

$$\widetilde{H}_t(\mathrm{Cl}(L_B(a))) \cong \widetilde{H}_{t-1}(\mathrm{Cl}(L_H(a))).$$

Proof. Put $K_a = \mathrm{Cl}(A[N_A(a)])$. If $a \in S_0$, its neighbourhood in B contains p, x but not y, and

$$\mathrm{Cl}(L_B(a)) = (p * K_a) \cup (x * X_a), \qquad X_a = \mathrm{Cl}(A[N_A(a) \cap S]).$$

Both pieces are cones and their intersection is X_a. The edge deletions defining H remove exactly the neighbours of a in T_0. Edges among its remaining neighbours in S are unchanged. Thus $X_a = \mathrm{Cl}(L_H(a))$, and Lemma 6.2 proves the assertion. The case $a \in T_0$ is obtained by exchanging the two profiles and auxiliaries.

If $a \in Z$, none of the edges incident to a is deleted. The neighbourhood contains both auxiliaries. Restrict the construction (12) to $N_A(a)$, writing

$$X_a = \mathrm{Cl}(A[N_A(a) \cap S]), \quad Y_a = \mathrm{Cl}(A[N_A(a) \cap T]), \quad W_a = X_a \cap Y_a.$$

The resulting complex V_a is acyclic by the same union-of-cones argument. Now $\mathrm{Cl}(L_B(a)) = (p * K_a) \cup V_a$, with intersection $X_a \cup Y_a$. The proof of (11), restricted to $N_A(a)$, identifies this intersection with $\mathrm{Cl}(L_H(a))$. Lemma 6.2 again applies. Augmented chains cover all possible empty neighbourhoods. □

The auxiliary neighbourhoods have the exact descriptions

$$\mathrm{Cl}(L_B(x)) = X \cup (y * W), \qquad \mathrm{Cl}(L_B(y)) = Y \cup (x * W). \tag{14}$$

They do not contain p.

Lemma 6.8. *If $U = X \cup Y$ is acyclic, the inclusions induce isomorphisms*

$$\widetilde{H}_t(W) \longrightarrow \widetilde{H}_t(X) \oplus \widetilde{H}_t(Y)$$

for all integers t.

Proof. For surjectivity, let a, b be cycles in X, Y. Since $a + b$ is a cycle in U, it bounds a chain $w = w_X + w_Y$ there. Then $z = a + \partial w_X = b + \partial w_Y$ is a cycle in W representing the prescribed pair of homology classes. For injectivity, suppose a cycle z in W bounds chains u in X and v in Y. The cycle $u + v$ bounds $w = w_X + w_Y$ in U. The chain $u + \partial w_X = v + \partial w_Y$ lies in W and has boundary z. These arguments apply in degree -1 as well, using the augmentation, and lower degrees are zero. □

When the kernel is acyclic, the auxiliary neighbourhoods are controlled by the opposite profile subgraphs.

Theorem 6.9. *If U is acyclic, then for every integer t,*

$$\widetilde{H}_t(\mathrm{Cl}(L_B(x))) \cong \widetilde{H}_{t-1}(Y), \qquad \widetilde{H}_t(\mathrm{Cl}(L_B(y))) \cong \widetilde{H}_{t-1}(X).$$

Proof. By Lemma 6.8, the map $\widetilde{H}_r(W) \to \widetilde{H}_r(X)$ is surjective and its kernel is isomorphic to $\widetilde{H}_r(Y)$: under the lemma's isomorphism it is the projection onto the first summand.

For $t \geq 0$, write a cycle of $X \cup (y * W)$ as $u + y * w$. The equations for being a cycle are $\partial w = 0$ and $\partial u = w$. Assign to it the class of w, which lies in the kernel of $\widetilde{H}_{t-1}(W) \to \widetilde{H}_{t-1}(X)$. The assignment respects boundaries: the boundary of $a + y * b$ changes w by ∂b. Every class in that kernel has a filling u in X, so the assignment is surjective.

If $w = \partial d$ in W, adding $\partial(y * d) = d + y * \partial d$ leaves the cycle $u + d$ in X. Surjectivity of $\widetilde{H}_t(W) \to \widetilde{H}_t(X)$ lets us replace this cycle, modulo a boundary in X, by a cycle in W. That cycle bounds in the cone $y * W$. The original cycle therefore bounds in $X \cup (y * W)$, proving injectivity. Its homology in degree -1 is zero because it contains y; the claimed right side then has degree -2 and is also zero. Lower degrees vanish. Use (14) and exchange X, Y for the second identity. □

6.5. **Equivalent finite conditions and their Euler form.** Here the layer construction of Section 3 has $d = 2$, $R = K_2$ and profiles S, T. We may therefore write the auxiliaries as x_i, y_i, and $|V(G_m)| = |V(A)| + 3m$. The identities just proved separate the obstructions on original vertices from those on the two auxiliary types.

Theorem 6.10. *Assume $S \cup T = V(A)$, $s \geq 0$, $A \in \mathcal{D}_s \setminus \mathcal{D}_{s-1}$, and $B - p \in \mathcal{D}_{s+1}$. Then the following are equivalent:*

1. *Every non-central neighbourhood $L_B(v)$, $v \neq p$, has non-trivial reduced homology.*
2. *The clique complex of the kernel H is acyclic, every $L_H(a)$, $a \in V(A)$, has non-trivial reduced homology, and both $A[S]$ and $A[T]$ have non-trivial reduced homology.*

Either condition implies

$$G_m \in \mathcal{D}_{s+m} \setminus \mathcal{D}_{s+m-1} \quad (m \geq 0).$$

Proof. The seed and remainder hypotheses imply acyclicity of K and $\mathrm{Cl}(B - p)$. Proposition 6.6 then implies acyclicity of $U = \mathrm{Cl}(H)$. Theorem 6.7 gives equivalence of the conditions on original vertices. Theorem 6.9 gives equivalence of those on the auxiliaries. Under condition (1), the data

satisfy precisely the homological criterion in Theorem 7.2, proved independently in Section 7. It applies to the displayed layered construction and proves strictness at every subsequent level. □

The remainder hypothesis is present on both sides of this equivalence. Acyclicity of the kernel is redundant under that hypothesis, but states explicitly the property needed when starting with a kernel. The equivalence does not dispense with the remainder hypothesis. Lemma 6.5 gives the covering condition on the profiles.

Extend q from graphs to finite complexes by putting $q(J) = \sum_{\sigma \in J} (-1)^{|\sigma|}$, including the empty face. This agrees with the existing notation $q(H) = q(\mathrm{Cl}(H))$. Inclusion–exclusion and the zero value on cones give identities without homological assumptions.

Proposition 6.11. *For the profile data above,*

$$q(B) = -q(H), \qquad q(B - p) = q(A) - q(H), \tag{15}$$

$$q(L_B(a)) = -q(L_H(a)) \qquad (a \in V(A)), \tag{16}$$

$$q(L_B(x)) = q(A[S]) - q(A[Z]), \qquad q(L_B(y)) = q(A[T]) - q(A[Z]). \tag{17}$$

If $q(H) = 0$, *the last line becomes*

$$q(L_B(x)) = -q(A[T]), \qquad q(L_B(y)) = -q(A[S]).$$

Proof. In (12), both pieces and their intersection are cones, so $q(V) = 0$. Apply inclusion–exclusion to (13) to obtain (15). The decompositions used to prove Theorem 6.7 likewise have pieces of zero q and intersection $\mathrm{Cl}(L_H(a))$, yielding (16). Apply inclusion–exclusion to (14) to obtain (17). Finally (11) gives $q(H) = q(A[S]) + q(A[T]) - q(A[Z])$. □

In particular, under $A \in \mathcal{D}_s \setminus \mathcal{D}_{s-1}$ and $B - p \in \mathcal{D}_s$, both have $q = 0$, so $q(H) = 0$. The non-zero Euler conditions on all non-central neighbourhoods of B are then equivalent to

$$q(L_H(a)) \neq 0 \quad \text{for all } a \in V(A), \qquad q(A[S]) \neq 0, \quad q(A[T]) \neq 0.$$

This is the kernel formulation of Theorem 3.6. The identities remain valid if the remainder is only assumed to lie in $\mathcal{D}_{s+1}$, as in its separate relaxation. The equality $q = 0$ is not used as a test for acyclicity or dismantlability.

One further necessary condition can be read from Lemma 6.8. If S, T, Z are non-empty and H is acyclic, the numbers of connected components satisfy

$$c(A[Z]) = c(A[S]) + c(A[T]) - 1.$$

Indeed, take dimensions in degree zero and use $\dim \widetilde{H}_0 = c - 1$. This numerical equality alone does not imply acyclicity; even agreement of dimensions in all degrees would not establish that the inclusion maps in the lemma are isomorphisms.

6.6. **Completing one edge and reducing the remainder.** The case of a single added edge lets us give an explicit reduction of the remainder. Suppose H is a graph, ab is a non-edge, and

$$A = H + ab, \qquad S = V(H) \setminus \{b\}, \qquad T = V(H) \setminus \{a\}.$$

Here $S_0 = \{a\}$ and $T_0 = \{b\}$, so deleting the edges between the exclusive profile parts recovers exactly H.

Lemma 6.12. *If* $\mathrm{Cl}(H)$ *is acyclic, inclusion induces*

$$\widetilde{H}_t(\mathrm{Cl}(L_H(v))) \cong \widetilde{H}_t(\mathrm{Cl}(H - v))$$

for every vertex v *and every integer* t.

Proof. Write $P = \mathrm{Cl}(H - v)$, $Q = v * \mathrm{Cl}(L_H(v))$ and $I = P \cap Q = \mathrm{Cl}(L_H(v))$. The union is acyclic by hypothesis, and Q is a cone. The same chain argument as in Lemma 6.8, now with the acyclic union $P \cup Q = \mathrm{Cl}(H)$, shows that $\widetilde{H}_t(I) \to \widetilde{H}_t(P) \oplus \widetilde{H}_t(Q)$ is an isomorphism. The second summand is zero. □

Thus, if H is acyclic and all its neighbourhoods have non-trivial reduced homology, then both $A[S] = H - b$ and $A[T] = H - a$ have non-trivial reduced homology. The remaining obligations for Theorem 6.10 are the strict level of the completed seed and the required level of $B - p$.

There is a useful sufficient condition for the latter. For disjoint graphs C, J, write $C \star J$ for their *complete join*.

Lemma 6.13. *If* $C \in \mathcal{D}_j$, $j \geq -1$, *then* $C \star J \in \mathcal{D}_j$ *for every graph* J.

Proof. For $j = -1$, an apex of C remains universal. Induct on j. Follow a level-j reduction of C. The neighbourhood of a deleted vertex in the join is the join of its current neighbourhood in C with J, and belongs to $\mathcal{D}_{j-1}$ by induction. When only one vertex of C remains, the join is a cone and reduces at level zero, hence at level j. This proves the assertion for $j \geq 0$. □

Proposition 6.14. *For the one-edge completion, put*

$$C = H[N_H(a) \cap N_H(b)].$$

The remainder $B - p$ *folds onto* $C \star C_4$, *where the cycle has vertices* a, b, y, x *in this order. Consequently, if* $C \in \mathcal{D}_j$ *for some* $j \geq 0$, *then* $B - p \in \mathcal{D}_j$.

Proof. Every $z \in V(H) \setminus \{a, b\}$ that is not adjacent to a in H is dominated by y in $B - p$. Indeed y is adjacent to every vertex except a, and a is not in the closed neighbourhood of z. Likewise, a vertex not adjacent to b is dominated by x. Delete all vertices outside $C \cup \{a, b, x, y\}$, retaining a, b, x, y and the common neighbours. Each chosen domination inclusion remains true after restricting to the vertices still present.

In the remaining graph, the edges on $\{a, b, x, y\}$ are exactly ab, by, yx, xa, and every vertex of C is adjacent to these four vertices. Its induced graph is unchanged from H. The remainder is therefore $C \star C_4$; when C is empty, it is just the four-cycle. If $C \in \mathcal{D}_j$, Lemma 6.13 gives $C \star C_4 \in \mathcal{D}_j$. Prepending the displayed folds is valid at level j by nesting of the classes. □

If C is a cone, the proposition applies with $j = 0$. The argument gives a sufficient certificate of the remainder. It does not assert that a failure to dismantle C at a chosen level excludes membership of $B - p$ at that level. In particular, an unsuccessful sufficient test must not replace the original remainder condition in the equivalence theorem.

6.7. **The completion underlying the eleven-vertex seed.** Define the completed seed A on $0, \ldots, 10$ by Table 12. It has thirty-three edges. The pair $\{1, 2\}$ is an edge; define $V(H) = V(A)$ and $E(H) = E(A) \setminus \{\{1, 2\}\}$. Then H has thirty-two edges and eleven vertices. For the profiles $S = V(A) \setminus \{2\}$ and $T = V(A) \setminus \{1\}$, the exclusive parts are $\{1\}$ and $\{2\}$, respectively. Hence this H is exactly the kernel of the profile data, and adding the single edge $\{1, 2\}$ to H gives A.

TABLE 12. The complete adjacency data of the completed seed A.

v	$N_A(v)$
0	$\{3,4,6,7,8\}$
1	$\{2,4,5,6,7,9,10\}$
2	$\{1,5,6,7,8,9,10\}$
3	$\{0,6,7,8,9,10\}$
4	$\{0,1,7,8,9,10\}$
5	$\{1,2,6,8,9\}$
6	$\{0,1,2,3,5,10\}$
7	$\{0,1,2,3,4,9\}$
8	$\{0,2,3,4,5,10\}$
9	$\{1,2,3,4,5,7\}$
10	$\{1,2,3,4,6,8\}$

This is the completed seed F_1 used in the article. The reductions of Appendix A establish its strict level and dismantle $B - p$. The homology calculation gives

$$\widetilde{H}_*(\mathrm{Cl}(H)) = 0, \qquad \dim_{\mathbb{F}_2} \widetilde{H}_1(\mathrm{Cl}(L_H(v))) = \begin{cases} 2, & v = 10, \\ 1, & v \neq 10, \end{cases}$$

with all other reduced homology groups of these neighbourhoods zero. Moreover, $N_H(1) \cap N_H(2) = \{5,6,7,9,10\}$ induces the path $10,6,5,9,7$. Proposition 6.14 therefore gives $B - p \in \mathcal{D}_0$. One such reduction begins with the folds $0,3,8 \to y$ followed by $4 \to x$, which leave the join of this path with the four-cycle on $1,2,y,x$. The displayed neighbourhood groups and Lemma 6.12 give condition (2) of Theorem 6.10. With the strict level of A proved in Appendix A, that theorem provides a homological proof of the strict levels of F_k.

Reproduce this calculation. From the repository directory, run

```
python reproduce.py -only kernels
```

The report reconstructs H by removing $\{1,2\}$, restores the seed by adding that edge, computes the displayed homology groups over $\mathbb{F}_2$, and checks the common-neighbour graph. It also compares the homology of each neighbourhood in H with the degree-shifted group in B.

7. HOMOLOGY THROUGH SUCCESSIVE LAYERS

The kernel calculations compare neighbourhoods in the first extension. We now follow those groups through every layer of the general construction, with any number of auxiliary types. As in Section 3, write $\mathcal{T} = \{0, \ldots, d-1\}$ for the type set. The coefficients and augmented chain conventions are those of Section 6. The degree shift will supply the criterion used in Theorem 6.10.

7.1. **A degree shift in every non-central neighbourhood.** For complexes on disjoint vertex sets, $K * J$ consists of unions of one face from each factor. If S^0 consists of two vertices without an edge, $K * S^0$ is the union of two cones on K. Lemma 6.2 gives

$$\widetilde{H}_t(K * S^0) \cong \widetilde{H}_{t-1}(K). \tag{18}$$

This includes $K = \{\varnothing\}$, for which the join is S^0. The clique complex of the complete join of two graphs is the join of their clique complexes, since a clique is a union of a clique in each graph.

Theorem 7.1. *For $m \geq 1$ and every integer t,*

$$\widetilde{H}_t\big(Cl(L_{G_m}(a))\big) \cong \widetilde{H}_{t-m+1}\big(Cl(L_B(a))\big), \qquad a \in V(A), \tag{19}$$

$$\widetilde{H}_t\big(Cl(L_{G_m}(x_{j,i}))\big) \cong \widetilde{H}_{t-m+1}\big(Cl(L_B(x_j))\big), \qquad j \in \mathcal{T}, \quad 0 \leq i < m. \tag{20}$$

Proof. First fix $a \in V(A)$ and put $\Lambda_m = L_{G_m}(a)$. Its vertices are $N_A(a)$, all centres, and all copies of the types j for which $a \in P_j$. For $m \geq 2$, delete a centre c_i from Λ_m. The folds $x_{j,h} \to x_{j,i}$, $h \neq i$, are still valid: restrict the neighbourhood inclusions in the proof of Lemma 3.2 to $N_A(a)$ and the types present. After these folds, the remaining graph is a cone with a remaining centre as apex. If there are no such auxiliary types, it is already a cone. Lemmas 6.1 and 6.3 imply that $Cl(\Lambda_m - c_i)$ is acyclic.

The neighbourhood of c_i inside Λ_m is isomorphic to Λ_{m-1}, using the same layer reindexing restricted to the vertices under consideration. Hence the decomposition

$$Cl(\Lambda_m) = Cl(\Lambda_m - c_i) \cup \big(c_i * Cl(L_{\Lambda_m}(c_i))\big)$$

has acyclic pieces and intersection isomorphic to $Cl(\Lambda_{m-1})$. Lemma 6.2 shifts homology by one degree. Iteration proves (19); at $m = 1$ it is the identity. The set $N_A(a)$ may be empty, but Λ_1 still contains p.

Now fix an auxiliary vertex $x_{j,i}$. Its neighbourhood has the vertices of P_j, all copies of types in $N_R(j)$, the vertices $x_{j,h}$ for $h \neq i$, and the centres c_h for $h \neq i$. For each $u \in N_R(j)$, fold $x_{u,h}$ onto $x_{u,i}$ for $h \neq i$. The source and target have the same adjacencies to P_j and auxiliary types. The target sees all centres in this neighbourhood, whereas the source may omit c_h. These folds therefore preserve homology, and their targets remain throughout.

What remains consists of a copy of $L_B(x_j)$ on

$$P_j \sqcup \{x_{u,i} : u \in N_R(j)\}$$

and the $m-1$ pairs $\{c_h, x_{j,h}\}$, $h \neq i$. Each pair has no internal edge. Distinct pairs are completely joined, and every pair is completely joined to the displayed copy of $L_B(x_j)$. The remaining clique complex is consequently

$$Cl(L_B(x_j)) * (S^0)^{*(m-1)}.$$

Apply Lemma 6.3 and then (18) $m-1$ times. If $L_B(x_j)$ is empty, both P_j and $N_R(j)$ are empty. There are then no folds, the first factor is $\{\varnothing\}$, and the same suspension identity applies. For $m = 1$ there are no pairs, so this case too is the identity. □

7.2. **The homological criterion.** Non-zero groups in the first layer remain non-zero after every shift. Together with a reduction of the remainder, they give the following criterion for strictness.

Theorem 7.2. *Let $s \geq 0$. Suppose that*

1. *$A \in \mathcal{D}_s \setminus \mathcal{D}_{s-1}$;*
2. *$B - p \in \mathcal{D}_{s+1}$;*
3. *for every $v \in V(B) \setminus \{p\}$, some integer t satisfies $\widetilde{H}_t(Cl(L_B(v))) \neq 0$.*

Then for every $m \geq 0$,

$$G_m \in \mathcal{D}_{s+m} \setminus \mathcal{D}_{s+m-1}, \qquad |V(G_m)| = |V(A)| + (d+1)m.$$

Proof. Theorem 7.1 propagates non-zero reduced homology to every non-central neighbourhood in every G_m, $m \geq 1$. By Lemma 6.4, none of these neighbourhoods belongs to any class $\mathcal{D}_j$.

The case $m = 0$ is the first hypothesis. For $m = 1$, the centre has neighbourhood $A \in \mathcal{D}_s$ and its deletion leaves $B - p \in \mathcal{D}_{s+1}$. Equation (1) yields $B \in \mathcal{D}_{s+1}$. A first deletion at level s is impossible: the centre's neighbourhood is not in $\mathcal{D}_{s-1}$, and neither is any non-central neighbourhood. Since B has more than one vertex, (1) gives $B \notin \mathcal{D}_s$. This includes $s = 0$, when $\mathcal{D}_{s-1}$ is the class of cones.

For $m \geq 2$, suppose G_{m-1} is strictly $(s + m - 1)$-dismantlable. Each centre has neighbourhood isomorphic to G_{m-1}; deleting it leaves a graph in $\mathcal{D}_0 \subseteq \mathcal{D}_{s+m}$ by Lemmas 3.1 and 3.2. Thus $G_m \in \mathcal{D}_{s+m}$. A first deletion at level $s + m - 1$ would require a neighbourhood in $\mathcal{D}_{s+m-2}$. Centres fail this condition by the induction hypothesis, and non-central vertices fail it by homology. This proves strictness. The order follows from (2). □

7.3. **Comparison with Euler characteristic.** The obstruction q of Section 3 is the alternating sum of reduced Betti numbers, with a contribution from degree -1. Rank–nullity in the augmented chain complex gives

$$q(H) = \sum_{r \geq 0} (-1)^r \dim_{\mathbb{F}_2} \widetilde{H}_{r-1}(\mathrm{Cl}(H)). \tag{21}$$

Indeed, if $f_t = \dim \widetilde{C}_t$, $b_t = \dim \widetilde{H}_t$, and $\rho_t = \operatorname{rank} \partial_t$, then $f_t = b_t + \rho_t + \rho_{t+1}$. In the finite alternating sum the ranks cancel, including the term of degree -1. In particular, $q(H) \neq 0$ implies non-trivial reduced homology.

The converse need not hold. The graph $C_4 \sqcup K_1$ has five vertices, four edges and no triangles, so $q = 1 - 5 + 4 = 0$. Its two components give $\dim \widetilde{H}_0 = 1$. The boundary matrix of the four cycle edges has rank three, and there are no two-dimensional faces, giving $\dim \widetilde{H}_1 = 1$. These contributions cancel in (21). This example separates the two obstructions on a graph; it does not by itself supply initial data satisfying all three hypotheses of Theorem 7.2.

The Euler criterion in Theorem 3.6 uses $q(L_B(v)) \neq 0$ for every non-central v, together with $A \in \mathcal{D}_s \setminus \mathcal{D}_{s-1}$ and the stronger remainder condition $B - p \in \mathcal{D}_s$. Equation (21) and Lemma 2.3 imply the hypotheses of Theorem 7.2. Allowing the remainder in $\mathcal{D}_{s+1}$ is a separate change from replacing Euler characteristic by homology: the first-deletion argument at $m = 1$ also permits this weaker remainder in the Euler proof. Both criteria retain a distinct requirement on the remainder; non-trivial homology of the other neighbourhoods alone does not ensure the positive reduction.

Reproduce this calculation. From the repository directory, run

```
python reproduce.py -only homology
```

The calculation checks the empty graph, a cone, $C_4 \sqcup K_1$, and an auxiliary with an empty first-layer neighbourhood. In the last example, after m layers its only non-zero reduced Betti number is $\widetilde{\beta}_{m-2} = 1$, for the tested values $m = 1, 2, 3$. The full command `python reproduce.py -extended` additionally checks all labelled seeds on one, two or three vertices, all profiles for one or two types, and all type graphs, at $m = 0, 1, 2, 3$. Homology is computed for the subcollection with seed order at most two. These are finite checks of the identities; Theorem 7.1 proves them for arbitrary seeds and all layers.

APPENDIX A. THE ARTICLE'S ELEVEN-VERTEX SEED AND ITS EXTENSION

The seed $F_1 = A$ is given in Table 12. Here we record the finite reductions and Euler obstructions that establish the strict levels of the $3k + 8$ family. Its profiles and type graph are

$$P_0 = V(A) \setminus \{2\}, \qquad P_1 = V(A) \setminus \{1\}, \qquad R = K_2. \tag{22}$$

The first extension $F_2 = B$ adds p, x, y, with p adjacent to every seed vertex and to neither auxiliary. The vertex x is adjacent to P_0, the vertex y to P_1, and xy is an edge. Thus A has eleven vertices and thirty-three edges, while B has fourteen vertices and sixty-five edges. Figure 1 shows the added edges in this first extension.

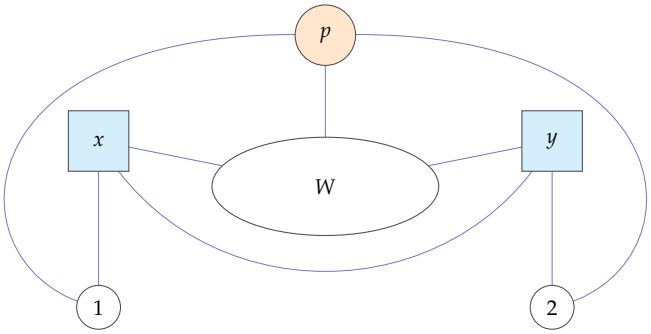


FIGURE 1. The thirty-two added edges. The ellipse W represents $V(A) \setminus \{1, 2\}$, the set Z of Section 6; an edge to W means adjacency to all nine vertices. Edges within A, including $\{1, 2\}$, are specified by Table 12.

A.1. **The seed is strictly 1-dismantlable.** We will use folds in two different graphs. To show that A belongs to level one, we fold each neighbourhood graph needed in a reduction. To exclude level zero, we examine the folds available in A itself.

Lemma A.1. *The graph A belongs to $\mathcal{D}_1$.*

Proof. We delete the vertices of A in the order

$$5, 2, 1, 4, 0, 6, 7, 8, 9, 3,$$

leaving 10. Each row of Table 13 justifies one deletion. The second column lists the neighbours of the vertex to be deleted, after the preceding deletions from A. The third column gives the folds in the graph induced by those neighbours, and the last column gives the surviving vertex.

To check a listed fold $u \to t$, let H be the neighbourhood graph for that row after its preceding folds. The adjacencies in Table 12 give $N_H[u] \subseteq N_H[t]$, so the fold is valid by Lemma 2.1. In the final two rows the neighbourhood graph is already a single vertex. Thus each row gives a reduction of its neighbourhood graph to one vertex, which puts that graph in $\mathcal{D}_0$ by Definition 2.2. The ten deletions therefore dismantle A at level one. □

TABLE 13. A complete level-one reduction of A.

Deleted vertex	Current neighbourhood	Folds within that neighbourhood	Survivor
5	$\{1,2,6,8,9\}$	$1 \to 2,\ 6 \to 2,\ 8 \to 2,\ 2 \to 9$	9
2	$\{1,6,7,8,9,10\}$	$6 \to 1,\ 7 \to 1,\ 8 \to 10,\ 9 \to 1,\ 1 \to 10$	10
1	$\{4,6,7,9,10\}$	$6 \to 10,\ 7 \to 4,\ 9 \to 4,\ 4 \to 10$	10
4	$\{0,7,8,9,10\}$	$9 \to 7,\ 7 \to 0,\ 0 \to 8,\ 8 \to 10$	10
0	$\{3,6,7,8\}$	$6 \to 3,\ 7 \to 3,\ 3 \to 8$	8
6	$\{3,10\}$	$3 \to 10$	10
7	$\{3,9\}$	$3 \to 9$	9
8	$\{3,10\}$	$3 \to 10$	10
9	$\{3\}$	none	3
3	$\{10\}$	none	10

Lemma A.2. *The graph A does not belong to $\mathcal{D}_0$.*

Proof. To determine whether a level-zero reduction can start, we list the folds available in A. By Lemma 2.1, a fold of u onto t in the current graph H is permitted exactly when $u \neq t$ and $N_H[u] \subseteq N_H[t]$. Comparing the closed neighbourhoods from Table 12 gives the complete lists below, first in A and then after each forced deletion:

H	all permitted folds
A	$5 \to 2$
$A - 5$	$9 \to 7$
$A - \{5,9\}$	none.

Any level-zero reduction would have to delete 5 and then 9. The remaining graph on $\{0,1,2,3,4,6,7,8,10\}$ admits no fold, so the reduction stops before reaching K_1. □

A.2. **A dismantlable remainder.** The neighbourhood graph of p is A, so p can be deleted at level two. Theorem 3.6 uses the hypothesis that the remaining graph $B - p$ can be dismantled at level one. In fact, ordinary folds already suffice.

Lemma A.3. *The graph $B - p$ belongs to $\mathcal{D}_0$.*

Proof. After removing p, the remaining vertices are $0, \ldots, 10, x, y$. Perform the following twelve folds in order:

$$\begin{aligned} &0 \to x, \quad 3 \to x, \quad 4 \to x, \quad 7 \to 9, \quad 8 \to y, \quad 9 \to 5, \\ &1 \to 6, \quad 2 \to 6, \quad 5 \to 6, \quad 6 \to 10, \quad 10 \to x, \quad x \to y. \end{aligned} \tag{23}$$

At each step, the adjacency table and the specified adjacencies of x, y give the closed-neighbourhood inclusion for the indicated fold in the graph left by the preceding folds. Only y survives. Thus $B - p \in \mathcal{D}_0 \subseteq \mathcal{D}_1$, giving the required hypothesis of Theorem 3.6 for $s = 1$. □

A.3. **Euler obstructions in the first extension.** A first deletion at level one would require a neighbourhood graph in $\mathcal{D}_0$. The centre fails this test because its neighbourhood graph is A. For each other vertex, we use Lemma 3.4: a non-zero q excludes its neighbourhood graph from $\mathcal{D}_0$.

Lemma A.4. *Every non-central vertex v of B satisfies $q(B[N_B(v)]) \neq 0$.*

Proof. For this clique table only, write $H_v = B[N_B(v)]$ for each of the thirteen vertices $v \neq p$. This local notation follows the article table and is unrelated to the family H_k of Section 4. To evaluate $q(H_v)$, we count the cliques of H_v by size. In Table 14, $c_r(H_v)$ counts the cliques of size r. The empty clique gives $c_0(H_v) = 1$, and none of the neighbourhood graphs has a clique of size greater than four.

To obtain a row directly from the adjacency data, enumerate the subsets of $N_B(v)$ and count those in which every pair of vertices is adjacent. These counts are the coefficients of C_{H_v}. Alternatively, compute the same polynomial using the recurrence in Lemma 3.3, starting from $C_\varnothing(z) = 1$. Each step uses graphs with fewer vertices, so the recursion terminates. In either case, evaluation at $z = -1$ gives the last column of Table 14, where every entry is non-zero. □

TABLE 14. All clique counts of $H_v = B[N_B(v)]$ for $v \neq p$.

v	$c_0(H_v)$	$c_1(H_v)$	$c_2(H_v)$	$c_3(H_v)$	$c_4(H_v)$	$q(H_v)$
0	1	8	21	20	5	−1
1	1	9	25	24	6	−1
2	1	9	24	21	4	−1
3	1	9	25	24	6	−1
4	1	9	25	24	6	−1
5	1	8	20	16	2	−1
6	1	9	24	21	4	−1
7	1	9	25	25	7	−1
8	1	9	24	21	4	−1
9	1	9	25	24	6	−1
10	1	9	25	23	4	−2
x	1	11	35	37	11	−1
y	1	11	35	36	10	−1

A.4. **The family at every strict level.** The checks on A and B now let us apply Theorem 3.6 to any number of layers, including zero layers.

Theorem A.5. *For every integer* $k \geq 1$, *the graph* $F_k = G_{k-1}$ *with* $d = 2$ *and* $R = K_2$, *specified by Table 12,* (22), *and the adjacency rules of Section 3 satisfies*

$$F_k \in \mathcal{D}_k \setminus \mathcal{D}_{k-1}, \qquad |V(F_k)| = 3k + 8.$$

Proof. Lemmas A.1 and A.2 establish $A \in \mathcal{D}_1 \setminus \mathcal{D}_0$. The second hypothesis follows from $B - p \in \mathcal{D}_0 \subseteq \mathcal{D}_1$ by Lemma A.3, and the last is exactly Lemma A.4.

Apply Theorem 3.6 with $s = 1$ and $d = 2$. It gives strict level $m + 1$ and order $11 + 3m$ after m layers, for every $m \geq 0$. Taking $m = k - 1$ gives level k and order $11 + 3(k - 1) = 3k + 8$. The first two graphs are $F_1 = A$ and $F_2 = B$. □

Reproduce this calculation. From the repository directory, run

```
python reproduce.py -only article
```

The report checks all ten outer deletions and twenty-three inner folds of the seed, the forced folds excluding level zero, the twelve folds of $B - p$, and all thirteen clique rows. It also checks the exact incidences drawn in the figures: the eight schematic incidences in Figure 1 expand to thirty-two added edges. Figure 2 displays the seed by its twenty-two missing edges; Figure 3 shows all thirty edges on the nine vertices added in three layers.

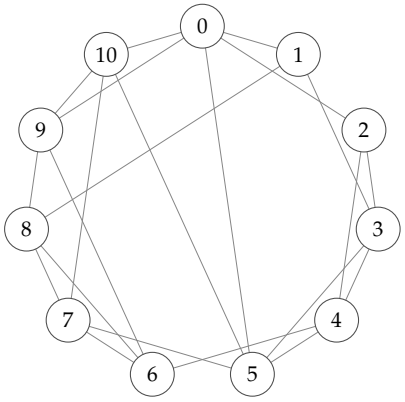


FIGURE 2. The complement of the eleven-vertex seed, with the twenty-two non-edges listed by the adjacency data.

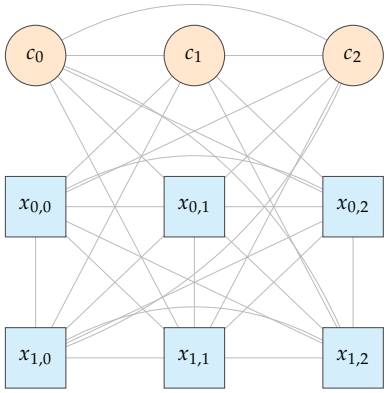


FIGURE 3. The new vertices in three layers. Each column contains its centre and two auxiliaries. The seed and its incident edges are omitted. A centre is adjacent to every other centre and to the auxiliaries outside its own column. All six auxiliaries are pairwise adjacent.

A.5. **Deleting one vertex.** Can a single vertex deletion preserve the strict level of F_k? The proof of strictness already provides two ways to examine the remaining graph. After deleting a centre we have an explicit fold reduction. For any other vertex, the Euler recurrence relates the value of q on the remaining graph to the value on its neighbourhood graph.

Proposition A.6. *Let $k \geq 2$ and let F_k be the graph of Theorem A.5. Deleting a centre gives a graph in $\mathcal{D}_0$. Deleting any other vertex gives a graph outside every class $\mathcal{D}_j$, $j \geq -1$. Consequently no single vertex deletion preserves its strict level k.*

Proof. Write $F_k = G_m$ with $m = k - 1 \geq 1$. If we delete a centre when $m = 1$, the remainder is $B - p$, which folds to one vertex by Lemma A.3. For $m \geq 2$, Lemma 3.2 gives a fold reduction after deleting any centre. In both cases the remainder belongs to $\mathcal{D}_0 \subseteq \mathcal{D}_{k-1}$ and thus cannot have strict level k.

Now let v be a vertex other than a centre. By Lemma 3.5, the non-zero values in Table 14 remain non-zero as we add layers, so $q(F_k[N_{F_k}(v)]) \neq 0$. The graph F_k itself belongs to $\mathcal{D}_k$ by Theorem A.5, so $q(F_k) = 0$ by Lemma 3.4. Using these two values in the recurrence of Lemma 3.3 gives

$$q(F_k - v) = q(F_k) + q(F_k[N_{F_k}(v)]) = q(F_k[N_{F_k}(v)]) \neq 0.$$

Lemma 3.4 therefore excludes $F_k - v$ from every dismantling class. □

The conclusion concerns one deletion from F_k, starting with $F_2 = B$. Minimality among all proper induced subgraphs and least order among all graphs of strict level k would require separate arguments.

Reproduce this calculation. From the repository directory, run

```
python reproduce.py -only deletions
```

The calculation checks every single-vertex deletion from F_2, F_3, F_4. It verifies the fold reduction after a centre deletion and the non-zero Euler value after any other deletion. Proposition A.6 proves the dichotomy for every $k \geq 2$.

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